

A conditional construction of a nonhyperlinear group and the centralizer problem

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ABSTRACT. We show that a positive answer to the centralizer problem for tracial matrix ultraproducts implies the existence of a finitely generated nonhyperlinear group.

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1. Introduction

In his 1976 classification paper, Connes suggested that every separable II_1 factor should embed into an ultrapower R^ω of the hyperfinite II_1 factor [7]. Kirchberg subsequently placed this question in a broad circle of equivalences involving tensor products and the QWEP problem for C^* -algebras [14]. Work relating the embedding problem to Tsirelson's problem connected it with finite-dimensional quantum correlations [13]. The theorem $\text{MIP}^* = \text{RE}$ of Ji, Natarajan, Vidick, Wright, and Yuen ultimately disproved the Connes embedding problem [12].

The corresponding group-theoretic problem remains open, while a recent breakthrough by OpenAI has produced the first example of a nonsofic group [19], see also [17]. Rădulescu introduced hyperlinear groups [21]: a countable group is hyperlinear if it embeds into the unitary group of a tracial ultraproduct of matrix algebras. Elek and Szabó proved that every sofic group is hyperlinear [10]. Although the general Connes embedding problem has a negative answer, no nonhyperlinear group is known.

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The study of sofic groups is in a sense more approachable, since it allows for input from finite combinatorics. Kun proved that every sofic approximation of a Kazhdan group is, up to a negligible modification, a disjoint union of expanders [15]. Building on this decomposition, Kun and the author showed that the sufficiently accurate almost automorphisms of such an expander sofic approximation form a group in a natural way [16]. Alekseev and the author subsequently proved that, after passing to an essentially equivalent finite model, the centralizer of a sofic embedding of a Kazhdan group in a universal sofic group is itself an internal metric ultraproduct of finite permutation groups [2]. Motivated by these results, they formulated the following matrix analogue [2, Open Problem 6.2(a)], which we call the centralizer problem.

THE CENTRALIZER PROBLEM.

Let G be a Kazhdan group and let $\pi: G \rightarrow U(\prod_{n \rightarrow \mathcal{U}} M_{d_n}(\mathbb{C}))$ be a homomorphism. Do there exist finite-dimensional $*$ -subalgebras $A_n \subseteq M_{d_n}(\mathbb{C})$ such that

$$\pi(G)' \cap \prod_{n \rightarrow \mathcal{U}} M_{d_n}(\mathbb{C}) = \prod_{n \rightarrow \mathcal{U}} A_n?$$

REMARK 1.1. The formulation in [2, Open Problem 6.2(a)] allows first replacing d_n by m_n , where $m_n/d_n \rightarrow_{\mathcal{U}} 1$. This flexibility is in fact unnecessary for the internality assertion above, by Lemma 2.3 below. This contrasts with the situation for sofic approximations and with the correction of approximate representations in normalized Hilbert–Schmidt norm, for which a change of dimension can be essential; see Becker and Lubotzky [4, Appendix A].

The purpose of this paper is to show that a positive answer to this problem implies the existence of a finitely generated nonhyperlinear group. The construction announced by OpenAI [19] inspired Kun and the author to isolate the mechanism used in the proof of [19, Proposition 2.3] in [17]. For a subgroup $H < G$, put $P_H = \{t \in G : tHt^{-1} \leq H\}$, the so-called *compression semigroup* of H in G . Following [17], we call H *infranormal* in G if P_H generates G . They proved that if $H < G$ are Kazhdan, H is infranormal and nonnormal, then both $G *_H (H \times \mathbb{Z}/2\mathbb{Z})$ and the group double $G *_H G$ are nonsofic [17]. The key representation-theoretic statement is that the centralizer of the image of H in a universal sofic group is normalized by the image of G ; see [17, Theorem 4.1].

We prove a matrix counterpart to this result, conditional on a positive answer to the centralizer problem.

THEOREM 1.2. Assume that the centralizer problem has a positive answer. Let $H < G$ be Kazhdan groups and suppose that H is infranormal in G . If $\pi: G \rightarrow U(\mathcal{M})$ is a homomorphism into a tracial matrix ultraproduct, then

$$\pi(g)(\pi(H)' \cap \mathcal{M})\pi(g)^{-1} = \pi(H)' \cap \mathcal{M}$$

for every $g \in G$.

The proof proceeds in two steps: we first prepare exact inclusions in the finite-dimensional level, secondly, we show that there is a phenomenon which we call *no-drift*. Put $A = \pi(H)' \cap \mathcal{M}$ and $D = \pi(G)' \cap \mathcal{M}$. Choose $t_1, \dots, t_m \in P_H$ such that $G = \langle H, t_1, \dots, t_m \rangle$, and set $u_\ell = \pi(t_\ell)$. Since $t_\ell H t_\ell^{-1} \leq H$, one immediately has $u_\ell^* A u_\ell \subseteq A$. The centralizer problem gives aligned internal models $D_n \subseteq A_n$,

with representatives $u_{\ell,n} \in D'_n$. At finite level, the preceding inclusion becomes the one-sided near inclusion $A_n^- = u_{\ell,n}^* A_n u_{\ell,n} \subset_{2,\varepsilon_n} A_n^+ = A_n$. The relative correction theorem replaces it by an exact inclusion $A_n^- \subseteq A_n^+$ while preserving the common copy of D_n .

The crucial finite object is the Bratteli matrix of this inclusion. It yields an asymptotic p.m.p. self-coupling of the set of matrix blocks, where a block is weighted by the trace of its unit. To show that this coupling is the graph of a p.m.p. automorphism, assign to a block $M_{r_i}(\mathbb{C}) \otimes 1_{s_i}$ the scale r_i/s_i . A D_n -conditional median gives a bounded transform with conditional expectation $1/2$, and the common copy of D_n allows the same median on the source and target sides. Its ultraproduct is invariant under the u_ℓ , hence belongs to D and equals $1/2$. The resulting concentration forces the scale quotient along the coupling to converge to 1. We then show that the edge multiplicity is 1 on asymptotically full mass; the marginal estimates separately match r_i and s_i and make repeated rows and columns negligible. Thus the coupling is concentrated on a measure-preserving permutation, which gives the reverse near inclusion and hence $u_\ell^* A u_\ell = A$.

The application is immediate once Theorem 1.2 is available.

THEOREM 1.3. *Assume that the centralizer problem has a positive answer. If $H < G$ is infranormal and not normal, and both groups are Kazhdan, then $G *_H G$ is not hyperlinear.*

Moreover, if q be a prime power and $r, d \geq 3$, put $R_+ = \mathbb{F}_q[x_1, \dots, x_d]$ and $R = \mathbb{F}_q[x_1^{\pm 1}, \dots, x_d^{\pm 1}]$, and set $H = \text{EL}_r(R_+)$ and $G = \text{EL}_r(R) \rtimes \text{SL}_d(\mathbb{Z})$, where $\text{SL}_d(\mathbb{Z})$ acts by monomial substitutions, then $H < G$ satisfies these assumptions.

The pair in Theorem 1.3 is the explicit residually finite Kazhdan pair constructed in [17, Theorem E].

This paper is organized as follows. Section 2 collects the necessary facts about inclusions of internal finite-dimensional algebras. Section 3 proves a relative correction theorem for one-sided near inclusions. Section 4 proves the no-drift theorem and explains why the common internal subalgebra is essential. Section 5 applies the theorem to normalization of Kazhdan centralizers and then to the conditional nonhyperlinearity of a group double.

2. Inclusions of internal algebras

Fix a nonprincipal ultrafilter $\mathcal{U}$ on $\mathbb{N}$. All matrix algebras carry their normalized traces. We write $\mathcal{M} = \prod_{n \rightarrow \mathcal{U}} (M_{d_n}(\mathbb{C}), \text{tr}_n)$ for the tracial ultraproduct and $\|x\|_2 = \text{tr}_n(x^*x)^{1/2}$ at the finite level. If $A_n \subseteq M_{d_n}(\mathbb{C})$ are unital $*$ -subalgebras, then $\prod_{n \rightarrow \mathcal{U}} A_n$ denotes their tracial ultraproduct inside $\mathcal{M}$. These are the usual metric ultraproducts of finite tracial von Neumann algebras; for general background on von Neumann algebra ultraproducts, see [1].

DEFINITION 2.1. A countable group Λ is *hyperlinear* if there are a nonprincipal ultrafilter $\mathcal{U}$, integers d_n , and an injective homomorphism $\Lambda \rightarrow U(\prod_{n \rightarrow \mathcal{U}} M_{d_n}(\mathbb{C}))$.

For finite-dimensional subalgebras $B, A \subseteq M_d(\mathbb{C})$, put

$$B \subset_{2,\varepsilon} A \iff \sup_{b \in B, \|b\| \leq 1} \text{dist}_2(b, A) \leq \varepsilon.$$

Let E_A and E_B be the trace-preserving conditional expectations, and put $\|T\|_{\infty,2} = \sup_{\|x\| \leq 1} \|T(x)\|_2$ for a linear map T . Then

$$(2.1) \quad B \subset_{2,\varepsilon} A \iff \|(1 - E_A)E_B\|_{\infty,2} \leq \varepsilon.$$

Indeed, E_A is the orthogonal projection onto A in L^2 , and the image under E_B of the ambient operator-norm unit ball is precisely the operator-norm unit ball of B . This is the classical notion of ε -containment in 2-norm, introduced by Murray and von Neumann [18] and developed by Christensen [6]. The conditional-expectation formulation in (2.1) is used systematically by Popa, Sinclair, and Smith [20].

The following observation will be used repeatedly.

LEMMA 2.2. *If $A_n, B_n \subseteq M_{d_n}(\mathbb{C})$ and $\Pi_{n \rightarrow \mathcal{U}} B_n \subseteq \Pi_{n \rightarrow \mathcal{U}} A_n$, then there are $\varepsilon_n \rightarrow_{\mathcal{U}} 0$ such that $B_n \subset_{2,\varepsilon_n} A_n$.*

PROOF. Otherwise, on a set belonging to $\mathcal{U}$, one can choose contractions $b_n \in B_n$ with $\text{dist}_2(b_n, A_n)$ bounded below. The element $(b_n)_{n \rightarrow \mathcal{U}}$ belongs to $\Pi_{\mathcal{U}} B_n$ but not to $\Pi_{\mathcal{U}} A_n$, a contradiction. $\square$

Replacing d_n by m_n with $m_n/d_n \rightarrow_{\mathcal{U}} 1$, adding or deleting a corner of normalized trace tending to zero, and conjugating by partial isometries on the remaining corner do not change the tracial ultraproduct. We refer to such changes as *stable negligible modifications*. They will always be accompanied by estimates showing that the modified and original unit balls have 2-Hausdorff distance tending to zero.

Once such a modification has been made, we identify the old and new ultraproducts and continue to use the symbols d_n , tr_n , and $\|\cdot\|_2$. Thus harmless changes of normalized trace caused by a dimension ratio tending to one will not be recorded repeatedly.

LEMMA 2.3. *Let $m_n/d_n \rightarrow_{\mathcal{U}} 1$. For every sequence of unital $*$ -subalgebras $A_n \subseteq M_{m_n}(\mathbb{C})$ there are unital $*$ -subalgebras $B_n \subseteq M_{d_n}(\mathbb{C})$ such that, under the canonical identification of the ambient tracial ultraproducts, $\Pi_{n \rightarrow \mathcal{U}} A_n = \Pi_{n \rightarrow \mathcal{U}} B_n$. Consequently, internality of a subalgebra is invariant under stable negligible changes of matrix dimension.*

PROOF. If the dimensions remain bounded along $\mathcal{U}$, the assumption implies $m_n = d_n$ on a set belonging to $\mathcal{U}$, and there is nothing to prove. We may therefore assume that the dimensions tend to infinity along $\mathcal{U}$. At indices where $m_n \leq d_n$, take $B_n = A_n \oplus \mathbb{C}1_{d_n - m_n}$. The added scalar corner has normalized trace tending to zero.

It remains to treat indices where $m_n > d_n$. Suppress the index n , put $k = m - d$, and write, after unitary conjugacy,

$$A = \bigoplus_{i=1}^t (M_{r_i}(\mathbb{C}) \otimes 1_{s_i}) \subseteq M_m(\mathbb{C}), \quad w_i = r_i s_i.$$

Choose j such that $S := \sum_{i < j} w_i \leq k < S + w_j$, and put $q = k - S$, $h = \min\{r_j, s_j\}$, and $q' = h[q/h]$. Then $0 \leq e := q' - q < h \leq \sqrt{w_j} \leq \sqrt{m}$. Delete the first $j - 1$ summands. In the j -th summand delete q' dimensions by replacing r_j with $r_j - q'/s_j$ if $s_j \leq r_j$, and by replacing s_j with $s_j - q'/r_j$ if $r_j < s_j$. Let p be the support projection of the retained algebra and put $C = pAp$, regarded as an algebra on

$p\mathbb{C}^m$. Then

$$\text{rank}(p) = m - S - q' = d - e, \quad \text{rank}(1 - p) = k + e = o_{\mathcal{U}}(m).$$

Write D_1 for the operator-norm unit ball of an algebra D and $d_{H,2}^{(m)}$ for Hausdorff distance in the normalized 2-norm of $M_m(\mathbb{C})$. Compression sends A_1 into C_1 up to an error supported on $1 - p$, while every element of C_1 has a norm-preserving block extension to A_1 . Hence

$$d_{H,2}^{(m)}(A_1, C_1) \leq 2\sqrt{\frac{k+e}{m}} = o_{\mathcal{U}}(1).$$

Let $Q \in M_m(\mathbb{C})$ be the standard projection of rank d and choose $R \leq Q$ of rank $d - e$. Since both p and R have codimension $k + e = o_{\mathcal{U}}(m)$, there is a unitary $u \in U(m)$ such that $upu^* = R$ and $\|u - 1\|_{2,m} = o_{\mathcal{U}}(1)$. Set

$$\tilde{B} = uCu^* \oplus \mathbb{C}(Q - R) \subseteq QM_m(\mathbb{C})Q \cong M_d(\mathbb{C}),$$

and let $B \subseteq M_d(\mathbb{C})$ be the corresponding algebra. The extra scalar corner has rank $e = o_{\mathcal{U}}(m)$, so the unit balls of A and B have normalized 2-Hausdorff distance tending to zero under the standard corner identification. Equality of the two ultraproduct algebras follows. Interchanging d_n and m_n gives the final assertion. $\square$

3. Correcting near inclusions

There is an extensive perturbation theory for finite von Neumann subalgebras. Besides the foundational work of Christensen [6], the closest general results to what follows are due to Popa, Sinclair, and Smith [20]. Under the symmetric hypothesis $\|E_A - E_B\|_{\infty,2} \leq \delta$, they obtain, for small δ , large cutdowns of A and B that are spatially isomorphic through a partial isometry close to the identity [20, Theorem 5.2]. They also prove that if $A \subseteq B$ is already an exact inclusion and $B \subset_{2,\delta} A$, with $\delta < 23^{-1/2}$, then A and B agree on a corner cut out by a projection $p \in Z(A' \cap B)$ satisfying $\text{tr}(p) \geq 1 - 23\delta^2$ [20, Theorem 3.5]. On the other hand, their one-sided example [20, Proposition 5.5 and the preceding discussion] shows that, for general finite von Neumann algebras, $B \subset_{2,\delta} A$ need not yield even a nonzero spatially embedded corner of B in A .

For a fixed finite-dimensional algebra, a related correction of approximate matrix units in uniform 2-norm is given by Carrión, Castillejos, Evington, Gabe, Schafhauser, Tikuisis, and White [5, Proposition 8.8]. Its tolerance depends on the chosen finite-dimensional source. It therefore does not give a uniform statement for a sequence B_n whose number and sizes of matrix blocks may grow.

What is needed here is consequently a specifically finite-dimensional and stable statement: the near inclusion is only one-sided, the algebras may vary without a complexity bound, a corner of asymptotically negligible dimension may be changed, and a common algebra D_n must be carried to the same corrected copy. These additional features are the content of Proposition 3.2 below, which is likely considered standard. We include the proof for completeness.

We use the following compact-group stability theorem of De Bondt and the author [8, Theorem 2.5].

THEOREM 3.1. *For every $L > 0$ and $\delta > 0$ there exists $\eta > 0$ with the following property. Let (K, d_K) be a compact metrizable group endowed with a compatible bi-invariant metric, and let $\varphi: K \rightarrow U(d)$ satisfy, for all $x, y \in K$,*

$$\|\varphi(xy) - \varphi(x)\varphi(y)\|_{2,d} \leq \eta \quad \text{and} \quad \|\varphi(x) - \varphi(y)\|_{2,d} \leq Ld_K(x, y) + \eta.$$

Then there exist an integer $m \in [d, (1 + \delta)d]$, a continuous homomorphism $\rho: K \rightarrow U(m)$, and an isometry $V: \mathbb{C}^d \rightarrow \mathbb{C}^m$ such that $\sup_{x \in K} \|\varphi(x) - V^ \rho(x) V\|_{2,d} < \delta$.*

A diagonal choice gives the ultrafilter form used below. If $\eta_n \rightarrow_{\mathcal{U}} 0$ and maps $\varphi_n: K_n \rightarrow U(d_n)$ satisfy the two estimates in the theorem with error η_n and a fixed L , then one may choose m_n , ρ_n , and V_n so that $m_n/d_n \rightarrow_{\mathcal{U}} 1$ and $\sup_{x \in K_n} \|\varphi_n(x) - V_n^* \rho_n(x) V_n\|_{2,d_n} \rightarrow_{\mathcal{U}} 0$. The theorem is the compact-group form of the method of de Chiffre, Ozawa, and the author [9, Theorem 5.2]; compare the unitary-group form in [3, Theorem 2.11]. The first result of this kind for the normalized 2-norm was proved by Gowers and Hatami for finite groups [11].

PROPOSITION 3.2. *Let $A_n, B_n \subseteq M_{d_n}(\mathbb{C})$ be unital finite-dimensional algebras and suppose $B_n \subset_{2,\varepsilon_n} A_n$, where $\varepsilon_n \rightarrow_{\mathcal{U}} 0$. After stable negligible modifications there are exact inclusions $\widehat{B}_n \subseteq \widehat{A}_n$ whose two ultraproducts agree with those of B_n and A_n , respectively. If $D_n \subseteq A_n \cap B_n$, the correction can be chosen so that the two identifications carry D_n to the same subalgebra on asymptotically full corners.*

PROOF. Write

$$B_n = \bigoplus_{i \in I_n} (M_{p_n,i}(\mathbb{C}) \otimes 1_{q_n,i}), \quad A_n = \bigoplus_{j \in J_n} (M_{r_n,j}(\mathbb{C}) \otimes 1_{s_n,j}).$$

Put $K_n = U(B_n) \times U(A'_n)$ and define $f_n: K_n \rightarrow U(d_n)$ by $f_n(b, a) = ba$. Equip K_n with the bi-invariant metric $d_{K_n}((b, a), (c, e)) = (\|b - c\|_2^2 + \|a - e\|_2^2)^{1/2}$. The maps f_n are $\sqrt{2}$ -Lipschitz. If $b \in U(B_n)$ and $a \in U(A'_n)$, the near inclusion gives $\|[b, a]\|_2 \leq 2\varepsilon_n$; hence the multiplicative defect of f_n is at most $2\varepsilon_n$.

Apply the ultrafilter form of Theorem 3.1 with $L = \sqrt{2}$ and $\eta_n = 2\varepsilon_n$. After replacing $\mathbb{C}^{d_n}$ by $\mathbb{C}^{m_n}$, where $m_n/d_n \rightarrow_{\mathcal{U}} 1$, there are continuous representations $\rho_n: K_n \rightarrow U(m_n)$ and isometries $V_n: \mathbb{C}^{d_n} \rightarrow \mathbb{C}^{m_n}$ such that

$$(3.1) \quad \sup_{x \in K_n} \|\rho_n(x) V_n - V_n f_n(x)\|_{2,m_n} \rightarrow_{\mathcal{U}} 0.$$

Indeed, fix $x \in K_n$ and put $c_n(x) = V_n^* \rho_n(x) V_n$. Expanding the square, using that $f_n(x)$ and $\rho_n(x)$ are unitary and that V_n is an isometry, gives

$$\begin{aligned} \|\rho_n(x) V_n - V_n f_n(x)\|_{2,m_n}^2 &= \frac{2d_n}{m_n} (1 - \Re \operatorname{tr}_n(f_n(x)^* c_n(x))) \\ &= \frac{2d_n}{m_n} \Re \operatorname{tr}_n(f_n(x)^* (f_n(x) - c_n(x))) \\ &\leq \frac{2d_n}{m_n} \|f_n(x) - c_n(x)\|_{2,d_n}. \end{aligned}$$

The last inequality is Cauchy–Schwarz, since $\|f_n(x)\|_{2,d_n} = 1$. The compression error tends to zero uniformly in x , and $d_n/m_n \rightarrow_{\mathcal{U}} 1$, which proves (3.1).

We use the following elementary observation using polar decomposition. If an exact intertwiner T_n is $o_2(1)$ from an isometry V_n , then its polar part W_n is also $o_2(1)$ from V_n and its initial and final complements have dimension $o_{\mathcal{U}}(d_n)$.

Indeed, the normalized dimension of the kernel is at most $\|T_n - V_n\|_2^2$, while singular-value comparison gives $\|W_n - T_n\|_2 \leq \|T_n - V_n\|_2$; the statement about the final complement follows from the dimension ratio tending to one.

The exact inclusion will come from restricting ρ_n to an appropriate K_n -invariant subspace of $\mathbb{C}^{m_n}$. We first define that subspace and then take the algebras associated with the two factor actions.

We now remove the irreducible summands of ρ_n that cannot occur in the two original factor actions. This is analogous to the passage to the standard isotypic component in [3, proof of Theorem 3.4]; here the near intertwiner in (3.1) gives the required dimension estimate directly.

Let $\sigma_n^B: U(B_n) \rightarrow U(d_n)$ and $\sigma_n^{A'}: U(A'_n) \rightarrow U(d_n)$ be the natural representations, so that $\sigma_n^B(b) = b$ and $\sigma_n^{A'}(a) = a$. The irreducible constituents of σ_n^B are the standard representations of the coordinate groups $U(p_{n,i})$, extended trivially over the other coordinates, and the constituent indexed by i occurs $q_{n,i}$ times. Similarly, the coordinate standard representation of $U(s_{n,j})$ occurs $r_{n,j}$ times in $\sigma_n^{A'}$.

Let P_n^B be the sum of the isotypic projections of $\rho_n|_{U(B_n) \times \{1\}}$ belonging to the irreducible constituents of σ_n^B . Define $P_n^{A'}$ in the same way from $\rho_n|_{\{1\} \times U(A'_n)}$ and $\sigma_n^{A'}$. Each of these projections is central in the finite-dimensional von Neumann algebra generated by its factor action. Since the two factor actions commute, P_n^B and $P_n^{A'}$ commute with $\rho_n(K_n)$ and with each other. Put $P_n^0 = P_n^B P_n^{A'}$ and take $\mathcal{H}_{0,n} = P_n^0 \mathbb{C}^{m_n}$. This space is K_n -invariant, so we may let $\rho_{0,n}$ be the restriction of ρ_n to $\mathcal{H}_{0,n}$ and define $\hat{B}_n = \rho_{0,n}(U(B_n), 1)''$ and $\hat{A}_n = \rho_{0,n}(1, U(A'_n))'$. The two factor actions commute, and hence $\hat{B}_n \subseteq \hat{A}_n$.

We next verify that $\mathcal{H}_{0,n}$ is asymptotically full. Define S_n^B and $S_n^{A'}$, both from $\mathbb{C}^{d_n}$ to $\mathbb{C}^{m_n}$, by

$$\begin{aligned} S_n^B &= \int_{U(B_n)} \rho_n(b, 1) V_n b^* db, \\ S_n^{A'} &= \int_{U(A'_n)} \rho_n(1, a) V_n a^* da. \end{aligned}$$

The first map is an exact $U(B_n)$ -intertwiner and the second is an exact $U(A'_n)$ -intertwiner. Equation (3.1) gives $\|S_n^B - V_n\|_{2,m_n} + \|S_n^{A'} - V_n\|_{2,m_n} \rightarrow_{\mathcal{U}} 0$. Their ranges are contained in $P_n^B \mathbb{C}^{m_n}$ and $P_n^{A'} \mathbb{C}^{m_n}$, respectively. The polar-decomposition observation therefore shows that each of these two spaces contains a subspace of dimension $d_n - o_{\mathcal{U}}(d_n)$ and that $\|(1 - P_n^B)V_n\|_{2,m_n} + \|(1 - P_n^{A'})V_n\|_{2,m_n} \rightarrow_{\mathcal{U}} 0$. P_n^B and $P_n^{A'}$ commute, so $\|(1 - P_n^0)V_n\|_{2,m_n} \rightarrow_{\mathcal{U}} 0$. It follows that $\dim P_n^0 \mathbb{C}^{m_n} \geq d_n - o_{\mathcal{U}}(d_n)$; the opposite bound $\dim P_n^0 \mathbb{C}^{m_n} \leq m_n = d_n + o_{\mathcal{U}}(d_n)$ gives $\dim \mathcal{H}_{0,n} = d_n + o_{\mathcal{U}}(d_n)$.

Every irreducible representation of the direct product K_n is a tensor product of irreducible representations of its two factors. By the definition of P_n^0 , only the coordinate standard representations can occur on $\mathcal{H}_{0,n}$. Thus there are integers $k_{n,ji} \geq 0$ such that

$$\mathcal{H}_{0,n} = \bigoplus_{i \in I_n, j \in J_n} (\mathbb{C}^{p_{n,i}} \otimes \mathbb{C}^{s_{n,j}})^{\oplus k_{n,ji}}$$

as a representation of $U(B_n) \times U(A'_n)$. These integers form the Bratteli matrix of the exact inclusion above. Put $r'_{n,j} = \sum_i k_{n,ji} p_{n,i}$ and $q'_{n,i} = \sum_j k_{n,ji} s_{n,j}$.

If no common algebra is specified, put $D_n = \mathbb{C}1$. Define

$$T_n^B = \int_{U(B_n)} \rho_n(b, 1) P_n^0 V_n b^* db,$$

$$T_n^A = \int_{U(D_n) \times U(A'_n)} \rho_n(d, a) P_n^0 V_n(da)^* d(d, a).$$

The first is an exact $U(B_n)$ -intertwiner and the second an exact $U(D_n) \times U(A'_n)$ -intertwiner. Equation (3.1) and the preceding support estimate imply that both are $o_2(1)$ from $P_n^0 V_n$, and hence from V_n . By the preceding observation, their polar parts W_n^B, W_n^A have initial and final complements of dimension $o_{\mathcal{U}}(d_n)$ and satisfy $\|W_n^B - V_n\|_2 + \|W_n^A - V_n\|_2 \rightarrow_{\mathcal{U}} 0$.

On their initial spaces, exact equivariance gives

$$(3.2) \quad \begin{aligned} W_n^B b &= \rho_{0,n}(b, 1) W_n^B \quad (b \in U(B_n)), \\ W_n^A da &= \rho_{0,n}(d, a) W_n^A \quad (d \in U(D_n), a \in U(A'_n)). \end{aligned}$$

After extension over complements of dimension $o_{\mathcal{U}}(d_n)$, the two partial isometries become unitaries between stably equivalent spaces with $\|W_n^B - W_n^A\|_2 \rightarrow_{\mathcal{U}} 0$. On their retained corners, both carry D_n , and every continuous functional calculus of a normal element of D_n , to the same copy $\rho_{0,n}(D_n, 1)$.

Comparing multiplicities gives

$$(3.3) \quad \frac{1}{d_n} \sum_j s_{n,j} |r'_{n,j} - r_{n,j}| + \frac{1}{d_n} \sum_i p_{n,i} |q'_{n,i} - q_{n,i}| \rightarrow_{\mathcal{U}} 0.$$

For example, $\sum_i p_{n,i} \min\{q'_{n,i}, q_{n,i}\} = d_n - o_{\mathcal{U}}(d_n)$, while both total dimensions are $d_n + o_{\mathcal{U}}(d_n)$; hence $\sum_i p_{n,i} |q'_{n,i} - q_{n,i}| = o_{\mathcal{U}}(d_n)$. The other estimate is identical. On the supports of the polar parts, the represented unitary groups agree exactly. If p_B, q_B are the initial and final projections of W_n^B , and p_A, q_A those of W_n^A , then exact equivariance gives

$$W_n^B(p_B B_n p_B)(W_n^B)^* = q_B \hat{B}_n q_B, \quad W_n^A(p_A A_n p_A)(W_n^A)^* = q_A \hat{A}_n q_A.$$

For the second identity, exact equivariance implies that p_A commutes with A'_n , and hence $p_A \in (A'_n)' = A_n$, while q_A commutes with $\rho_{0,n}(1, A'_n)$, and hence $q_A \in \hat{A}_n$. Taking commutants inside the two support corners gives $(p_A A'_n p_A)'_{p_A M_{d_n} p_A} = p_A A_n p_A$, and analogously on the corrected side. The discarded initial and final corners have normalized trace $o_{\mathcal{U}}(1)$, so extending contractions by zero changes their 2-norm by $o_{\mathcal{U}}(1)$, uniformly on the unit ball. Haar expectation and the Russo–Dye description of the unit ball therefore give

$$(3.4) \quad d_{H,2}((B_n)_1, (\hat{B}_n)_1) + d_{H,2}((A_n)_1, (\hat{A}_n)_1) \rightarrow_{\mathcal{U}} 0$$

after the stable identifications. This proves the equality of ultraproducts. Here $d_{H,2}$ denotes Hausdorff distance between operator-norm unit balls for the normalized 2-norm. The D_n -equivariance in (3.2) and the common extensions constructed above proves that D_n remained unchanged. $\square$

4. The no-drift argument

We now isolate the operator-algebraic core of the argument. After correcting a one-sided near inclusion, its Bratteli matrix, weighted by the normalized dimensions of the source–target cells, defines an asymptotic p.m.p. self-coupling of the finite

set of blocks. The conditional median relative to D_n shows that this coupling is concentrated on the graph of a p.m.p. automorphism. The resulting matching gives the reverse near inclusion. Remark 4.2 shows what can happen without D_n .

THEOREM 4.1. *Let $A = \prod_{n \rightarrow \mathcal{U}} A_n$ and $D = \prod_{n \rightarrow \mathcal{U}} D_n$ be internal finite-dimensional subalgebras of a tracial matrix ultraproduct, with $D_n \subseteq A_n$. Let $u_1, \dots, u_m \in U(\mathcal{M})$ have representatives $u_{\ell,n} \in D'_n$. Assume $u_\ell^* A u_\ell \subseteq A$ for $1 \leq \ell \leq m$ and $D = A \cap \bigcap_{\ell=1}^m \{u_\ell\}'$. Then $u_\ell^* A u_\ell = A$ for $1 \leq \ell \leq m$.*

PROOF. Write $A_n = \bigoplus_{i \in I_n} (M_{r_{n,i}}(\mathbb{C}) \otimes 1_{s_{n,i}})$, and let $z_{n,i}$ denote the corresponding minimal central projections. Define the positive central scale operator $\Delta_n = \sum_{i \in I_n} (r_{n,i}/s_{n,i}) z_{n,i} \in Z(A_n)_+$. On the i -th block, $r_{n,i}$ is the matrix size and $s_{n,i}$ is its multiplicity, so Δ_n records their ratio. In particular, Δ_n commutes with D_n . Write $\beta_S(T) = T(T+S)^{-1}$ for commuting positive invertible operators. If e is a minimal central projection of D_n , then $e\Delta_n(\Delta_n+t)^{-1}$ commutes with D_n , so bimodularity gives $E_{D_n}(e\Delta_n(\Delta_n+t)^{-1}) = c_{n,e}(t)e$. Indeed, the conditional expectation belongs to eD_n and commutes with eD_n , so it is scalar because e is minimal central. The scalar $c_{n,e}(t)$ is continuous and decreasing, and it tends to 1 as $t \downarrow 0$ and to 0 as $t \rightarrow \infty$. Choose $m_{n,e} > 0$ with $c_{n,e}(m_{n,e}) = 1/2$, put $m_n = \sum_e m_{n,e} e \in Z(D_n)_+$, and set $x_n = \beta_{m_n}(\Delta_n)$. Summing over the minimal central projections e gives $E_{D_n}(x_n) = \frac{1}{2}1$. We refer to m_n as the D_n -conditional median of Δ_n : on each minimal central summand of D_n , it balances the bounded transform at $1/2$. Indeed, $2x_n - 1 = (\Delta_n - m_n)(\Delta_n + m_n)^{-1}$, and the conditional expectation of this operator onto D_n is zero.

Fix ℓ and set $A_n^+ = A_n$ and $A_n^- = u_{\ell,n}^* A_n u_{\ell,n}$. By Lemma 2.2 and the assumed one-sided inclusion, $A_n^- \subset_{2,\varepsilon_n} A_n^+$ for some $\varepsilon_n \rightarrow_{\mathcal{U}} 0$. Because $u_{\ell,n} \in D'_n$, one has $D_n \subseteq A_n^+ \cap A_n^-$. Apply the relative form of Proposition 3.2, and denote the resulting exact inclusion by $\hat{A}_n^- \subseteq \hat{A}_n^+$. For the finite-dimensional calculation, suppress n from the block indices. Since A_n^- is conjugate to A_n^+ , use the same parameters r_i, s_i for both, with the source labels transported by $u_{\ell,n}$.

Let (k_{ji}) be the Bratteli matrix of the corrected inclusion, with source blocks indexed by i and target blocks indexed by j . Thus k_{ji} is the multiplicity with which the i -th source matrix block occurs inside the j -th target matrix block, and

$$\mathcal{H}_{0,n} = \bigoplus_{j,i \in I_n} (\mathbb{C}^{r_i} \otimes \mathbb{C}^{s_j})^{\oplus k_{ji}}.$$

Put $r'_j = \sum_i k_{ji} r_i$ and $s'_i = \sum_j k_{ji} s_j$. There is no reason for the Bratteli matrix to be diagonal. It also defines a correspondence between the central blocks of the source and target. The original block i has trace weight $r_i s_i / d_n$. The joint block (j, i) has dimension $k_{ji} r_i s_j$, so, with $\hat{d}_n = \dim \mathcal{H}_{0,n}$, put $\mu_n(j, i) = k_{ji} r_i s_j / \hat{d}_n$. This is a probability measure whose source and target marginals are $r_i s'_i / \hat{d}_n$ and $r'_j s_j / \hat{d}_n$. The marginal estimates below identify both asymptotically with the original block weights. Thus the weighted Bratteli matrix defines an asymptotic p.m.p. self-coupling of the block space, not yet the graph of a transformation.

The rest of the proof shows that this coupling is asymptotically the graph of a p.m.p. automorphism. The trace weight of a block records the product $r_i s_i$, whereas the scale operator records the ratio r_i / s_i . Equation (4.2) below shows that on a joint block the quotient of the corrected target and source scales is

$(r'_j/r_i)(s'_i/s_j) \geq k_{ji}^2$. The conditional median forces this quotient to converge to 1 in μ_n -measure. Hence $k_{ji} = 1$ and both factors converge to 1 off a set of negligible mass. The marginal estimates recover $r_j/r_i \rightarrow_{\mathcal{U}} 1$ and $s_i/s_j \rightarrow_{\mathcal{U}} 1$ and eliminate repeated rows and columns, leaving a partial permutation of asymptotically full mass. Let us explain this in more detail: for fixed i , the corresponding part of $\mathcal{H}_{0,n}$ is

$$\bigoplus_j (\mathbb{C}^{r_i} \otimes \mathbb{C}^{s_j})^{\oplus k_{ji}} \cong \mathbb{C}^{r_i} \otimes \mathbb{C}^{s'_i}.$$

Thus the i -th source block is $M_{r_i}(\mathbb{C}) \otimes 1_{s'_i}$. For fixed j , one has

$$\bigoplus_i (\mathbb{C}^{r_i} \otimes \mathbb{C}^{s_j})^{\oplus k_{ji}} \cong \mathbb{C}^{r'_j} \otimes \mathbb{C}^{s_j}.$$

Thus the j -th target block is $M_{r'_j}(\mathbb{C}) \otimes 1_{s_j}$. Consequently,

$$\hat{A}_n^- = \bigoplus_{i: s'_i > 0} (M_{r_i}(\mathbb{C}) \otimes 1_{s'_i}), \quad \hat{A}_n^+ = \bigoplus_{j: r'_j > 0} (M_{r'_j}(\mathbb{C}) \otimes 1_{s_j}).$$

On the source side the generated algebra retains r_i and changes the multiplicity s_i . On the target side taking the commutant retains s_j and changes the matrix size r_j .

Let $\widehat{\text{tr}}_n$ be the normalized trace on $\mathcal{H}_{0,n}$. Equation (3.3) now becomes

$$(4.1) \quad \frac{1}{d_n} \sum_j s_j |r'_j - r_j| + \frac{1}{d_n} \sum_i r_i |s'_i - s_i| \rightarrow_{\mathcal{U}} 0.$$

The first sum is the target dimension discrepancy and the second is the source dimension discrepancy. Moreover, $\hat{d}_n = \sum_j r'_j s_j = \sum_i r_i s'_i$. It follows from either half of (4.1) that $\hat{d}_n/d_n \rightarrow_{\mathcal{U}} 1$. Let $\hat{w}_i \in Z(\hat{A}_n^-)$ and $\hat{z}_j \in Z(\hat{A}_n^+)$ be the source and target block projections. The corrected scale operators are

$$\Delta_n^- = \sum_i \frac{r_i}{s'_i} \hat{w}_i, \quad \Delta_n^+ = \sum_j \frac{r'_j}{s_j} \hat{z}_j.$$

Zero corrected blocks are omitted from these sums and from the ratios below. The projections $\hat{w}_i$ commute with the projections $\hat{z}_j$: indeed, $\hat{A}_n^- \subseteq \hat{A}_n^+$ and $\hat{z}_j$ is central in $\hat{A}_n^+$. Their product is nonzero exactly when $k_{ji} > 0$, and then its range has dimension $k_{ji} r_i s_j$. On this joint block,

$$(4.2) \quad \frac{r'_j/s_j}{r_i/s'_i} = \frac{r'_j s'_i}{r_i s_j} \geq k_{ji}^2.$$

Here $r'_j \geq k_{ji} r_i$ and $s'_i \geq k_{ji} s_j$. Since k_{ji} is a positive integer on a nonzero joint block, the quotient is at least one. Thus $\Delta_n^- \leq \Delta_n^+$ on $\mathcal{H}_{0,n}$.

We next compare the original and corrected bounded scale operators. The original source scale is $u_{\ell,n}^* \Delta_n u_{\ell,n}$. Since $u_{\ell,n}$ commutes with m_n , its bounded transform is $y_n := u_{\ell,n}^* x_n u_{\ell,n} = \beta_{m_n}(u_{\ell,n}^* \Delta_n u_{\ell,n})$. Thus $x_n \in A_n^+$ and $y_n \in A_n^-$ are the uncorrected target and source bounded scales.

The relative correction gives a common $*$ -representation $\pi_{D,n}: D_n \rightarrow \hat{A}_n^- \subseteq \hat{A}_n^+$. Put $\widehat{m}_n = \pi_{D,n}(m_n)$. Let W_n^- be the relative intertwiner from A_n^- to $\hat{A}_n^-$ and let W_n^+ be the relative intertwiner from A_n^+ to $\hat{A}_n^+$. On their retained supports they satisfy $W_n^- m_n = \widehat{m}_n W_n^-$ and $W_n^+ m_n = \widehat{m}_n W_n^+$. Set $x_n^\pm = \beta_{\widehat{m}_n}(\Delta_n^\pm)$. The three operators Δ_n^- , Δ_n^+ , and $\widehat{m}_n$ commute: Δ_n^- is central in $\hat{A}_n^-$, $\widehat{m}_n$ belongs to $\hat{A}_n^-$,

and Δ_n^+ is central in $\widehat{A}_n^+$, which contains $\widehat{A}_n^-$. Since $t \mapsto t(t+s)^{-1}$ is increasing for $s > 0$, (4.2) gives $x_n^- \leq x_n^+$.

Exact equivariance on the retained supports gives

$$\begin{aligned} W_n^- y_n &= \beta_{\widehat{m}_n} \left(\sum_i \frac{r_i}{s_i} \widehat{w}_i \right) W_n^-, \\ W_n^+ x_n &= \beta_{\widehat{m}_n} \left(\sum_j \frac{r_j}{s_j} \widehat{z}_j \right) W_n^+. \end{aligned}$$

For the first identity, W_n^- intertwines the source algebra. For the second, W_n^+ intertwines D_n and $(A_n^+)'$; this is sufficient because $\Delta_n \in Z(A_n^+) \subseteq (A_n^+)'$ and $m_n \in D_n$. In Proposition 3.2, W_n^- and W_n^+ are the polar parts of exact intertwiners that are $o_2(1)$ from the same isometry. Their initial and final complements therefore have dimension $o_{\mathcal{U}}(d_n)$. Add a complementary space to the smaller of the original and corrected spaces, put $d_n^{\circ} = \max\{d_n, \widehat{d}_n\}$, and complete both polar parts to unitaries between these spaces of common dimension. Write tr_n° and $\|\cdot\|_{p,\circ}$ for the resulting normalized trace and norms. The completed unitaries satisfy $\|W_n^- - W_n^+\|_{2,\circ} \rightarrow_{\mathcal{U}} 0$ and $d_n^{\circ}/d_n \rightarrow_{\mathcal{U}} 1$. Since $\widehat{d}_n/d_n \rightarrow_{\mathcal{U}} 1$, also $\widehat{d}_n/d_n^{\circ} \rightarrow_{\mathcal{U}} 1$. On the added summands, extend x_n, y_n and the bounded transforms on the corrected space by zero.

The scalar estimate $|a/(a+s) - b/(b+s)| \leq \min\{1, |\log(a/b)|\}$ is uniform in $s > 0$. Indeed, after writing $a = e^{\alpha}$ and $b = e^{\beta}$, the derivative of $e^t/(e^t + s)$ is at most $1/4$, while both values lie in $[0, 1]$. Put $g(t) = \min\{1, |\log t|\}$ for $t > 0$ and $g(0) = 1$. The elementary bound $g(t)^2 \leq 2|t - 1|$ for $t \geq 0$ and (4.1) give

$$(4.3) \quad \sum_i \frac{r_i s_i}{d_n} g(s'_i/s_i)^2 + \sum_j \frac{r_j s_j}{d_n} g(r'_j/r_j)^2 \rightarrow_{\mathcal{U}} 0.$$

For example, the first sum is bounded by $2d_n^{-1} \sum_i r_i |s'_i - s_i|$; the second estimate is the same with the target indices.

Before completion, the source intertwiner retains on block i a corner of dimension at most $r_i \min\{s_i, s'_i\} \leq r_i s_i$. Indeed, its support is the identity on $\mathbb{C}^{r_i}$ tensored with a projection in each multiplicity space, and the initial and final supports of a partial isometry have the same rank. That rank is therefore at most both $r_i s_i$ and $r_i s'_i$. Similarly, the target intertwiner retains on block j a corner of dimension at most $s_j \min\{r_j, r'_j\} \leq r_j s_j$. Because the correction is relative to D_n , these support projections commute with the corresponding copies of D_n , and hence with m_n and $\widehat{m}_n$.

On a retained source block the two scale values are r_i/s_i and r_i/s'_i ; on a retained target block they are r_j/s_j and r'_j/s_j . The scalar estimate, applied pointwise to the spectrum of the commuting operator $\widehat{m}_n$, therefore gives

$$(4.4) \quad \begin{aligned} \|x_n^- W_n^- - W_n^- y_n\|_{2,\circ}^2 &\leq \frac{1}{d_n^{\circ}} \sum_i r_i s_i g(s'_i/s_i)^2 + o_{\mathcal{U}}(1), \\ \|x_n^+ W_n^+ - W_n^+ x_n\|_{2,\circ}^2 &\leq \frac{1}{d_n^{\circ}} \sum_j r_j s_j g(r'_j/r_j)^2 + o_{\mathcal{U}}(1). \end{aligned}$$

The $o_{\mathcal{U}}(1)$ terms come from the complementary summands added when the partial isometries were completed. Their normalized dimensions tend to zero, and all

bounded transforms involved are contractions. The two right-hand sides tend to zero by (4.3) and $d_n^\circ/d_n \rightarrow_{\mathcal{U}} 1$. The scalar estimate is uniform in the median parameter, so no bound on m_n or m_n^{-1} , equivalently on $\widehat{m}_n$ or $\widehat{m}_n^{-1}$, uniform in n is required.

We spell out how this order returns to the original space. On the common corrected space, Equation (4.4) says that $W_n^- y_n (W_n^-)^*$ is close to x_n^- and $W_n^+ x_n (W_n^+)^*$ is close to x_n^+ in normalized 2-norm. Since $x_n^- \leq x_n^+$ and $\|\cdot\|_{1,\circ} \leq \|\cdot\|_{2,\circ}$, the positive-part trace satisfies

$$\begin{aligned} & \text{tr}_n^\circ \left(\left(W_n^- y_n (W_n^-)^* - W_n^+ x_n (W_n^+)^* \right)_+ \right) \\ & \leq \|W_n^- y_n (W_n^-)^* - x_n^-\|_{1,\circ} + \|W_n^+ x_n (W_n^+)^* - x_n^+\|_{1,\circ} \rightarrow_{\mathcal{U}} 0. \end{aligned}$$

Here we used the elementary inequality $\text{tr}((a-b)_+) \leq \|a-c\|_1 + \|b-d\|_1$ when $c \leq d$. Since x_n is a contraction, $\|W_n^- x_n (W_n^-)^* - W_n^+ x_n (W_n^+)^*\|_{2,\circ} \leq 2\|W_n^- - W_n^+\|_{2,\circ} \rightarrow_{\mathcal{U}} 0$. Hence, after conjugating by W_n^- , the stabilized trace of $(y_n - x_n)_+$ tends to zero. Here we also use $|\text{tr}(a_+) - \text{tr}(b_+)| \leq \|a-b\|_1 \leq \|a-b\|_2$ for self-adjoint a, b when replacing the W_n^+ -transport of x_n by its W_n^- -transport. The stabilized trace is $(d_n/d_n^\circ) \text{tr}_n((y_n - x_n)_+)$, so $\text{tr}_n((y_n - x_n)_+) \rightarrow_{\mathcal{U}} 0$.

Since x_n and y_n have the same trace, the positive and negative parts of $y_n - x_n$ have equal trace. The operators x_n and y_n are positive contractions, so $\|y_n - x_n\|_{2,d_n}^2 \leq \|y_n - x_n\|_{1,d_n} = 2 \text{tr}_n((y_n - x_n)_+) \rightarrow_{\mathcal{U}} 0$.

This holds for every ℓ . Since each x_n belongs to A_n and $u_{\ell,n}^* x_n u_{\ell,n} - x_n \rightarrow_{\mathcal{U}} 0$ in 2-norm, the element $x = (x_n)_{n \rightarrow \mathcal{U}}$ belongs to $A \cap \bigcap_{\ell=1}^m \{u_\ell\}'$, and hence $x \in D$. Since D is internal, its conditional expectation is coordinatewise. Therefore $E_D(x) = (E_{D_n}(x_n))_{n \rightarrow \mathcal{U}} = \frac{1}{2}1$. As $x \in D$, this gives $x = E_D(x) = \frac{1}{2}1$. Hence x_n and y_n converge to $\frac{1}{2}$ in 2-norm. Equation (4.4) gives the same conclusion for x_n^- and x_n^+ after transport to the corrected space. For example, on the stabilized space,

$$\left\| x_n^- - \frac{1}{2} \right\|_{2,\circ} \leq \|x_n^- - W_n^- y_n (W_n^-)^*\|_{2,\circ} + \left\| y_n - \frac{1}{2} \right\|_{2,d_n} + o_{\mathcal{U}}(1).$$

The last term is the contribution of the stabilizing corner, whose relative dimension tends to zero. The same estimate holds for x_n^+ . Since $\widehat{d}_n/d_n^\circ \rightarrow_{\mathcal{U}} 1$, both corrected bounded scales converge to $\frac{1}{2}$ in the normalized 2-norm on $\mathcal{H}_{0,n}$.

It remains to turn this concentration into the claimed matching for the probability correspondence μ_n . Equation (4.1) shows that its marginals have total variation $o_{\mathcal{U}}(1)$ from the original block weights $r_i s_i / d_n$ and $r_j s_j / d_n$. For example, on the source side the change in the numerators is $\widehat{d}_n^{-1} \sum_i r_i |s'_i - s_i|$, while replacing the denominator $\widehat{d}_n$ by d_n costs $o_{\mathcal{U}}(1)$.

The operators Δ_n^- , Δ_n^+ , and $\widehat{m}_n$ commute and are invertible on every retained block. On their joint spectrum,

$$\frac{\Delta_n^+}{\Delta_n^-} = \frac{x_n^+ / (1 - x_n^+)}{x_n^- / (1 - x_n^-)} \rightarrow_{\mathcal{U}} 1 \quad \text{in } \mu_n\text{-measure.}$$

Indeed, $x_n^\pm / (1 - x_n^\pm) = \Delta_n^\pm \widehat{m}_n^{-1}$, so the common factor $\widehat{m}_n^{-1}$ cancels in the quotient. Given $\varepsilon > 0$, choose $\delta > 0$ so that the quotient of the two odds ratios is within ε of 1 whenever both arguments are within δ of $1/2$. Chebyshev's inequality and the 2-norm convergence of $x_n^\pm$ show that the exceptional trace tends to zero. The joint

block satisfies $\widehat{\text{tr}}_n(\widehat{z}_j \widehat{w}_i) = \mu_n(j, i)$, and the quotient on the left is constant there. By (4.2), on the joint block (j, i) it is the product $(r'_j/r_i)(s'_i/s_j)$, and each factor is at least k_{ji} . A joint block with $k_{ji} \geq 2$ therefore has quotient at least 4, so the union of all such blocks has μ_n -mass $o_{\mathcal{U}}(1)$. Equivalently, $\mu_n(\{(j, i) : k_{ji} \neq 1\}) \rightarrow_{\mathcal{U}} 0$. Moreover, each of the two factors lies between 1 and the quotient on every remaining joint block. It follows that $r'_j/r_i \rightarrow_{\mathcal{U}} 1$ and $s'_i/s_j \rightarrow_{\mathcal{U}} 1$ in μ_n -measure.

The marginal-error estimate gives the two identities

$$\begin{aligned} \sum_{j,i: k_{ji} > 0} \mu_n(j, i) \left| \frac{r_j}{r'_j} - 1 \right| &= \frac{1}{\widehat{d}_n} \sum_j s_j |r'_j - r_j| = o_{\mathcal{U}}(1), \\ \sum_{j,i: k_{ji} > 0} \mu_n(j, i) \left| \frac{s_i}{s'_i} - 1 \right| &= \frac{1}{\widehat{d}_n} \sum_i r_i |s'_i - s_i| = o_{\mathcal{U}}(1). \end{aligned}$$

Combining the corrected ratios with these two marginal ratios gives $r_j/r_i = (r_j/r'_j)(r'_j/r_i) \rightarrow_{\mathcal{U}} 1$ and $s_i/s_j = (s_i/s'_i)(s'_i/s_j) \rightarrow_{\mathcal{U}} 1$ in μ_n -measure.

Choose $\eta_n \rightarrow_{\mathcal{U}} 0$ and a set F_n of μ_n -measure tending to one such that, on F_n , $k_{ji} = 1$ and both last ratios lie between $1 - \eta_n$ and $1 + \eta_n$. We must still remove repeated rows and columns. Suppose first that a target j is incident to two distinct edges $(j, i_1), (j, i_2)$ of F_n . For each of them, $r_{i_a} \geq r_j/(1 + \eta_n) \geq (1 - \eta_n)r_j$, where $a = 1, 2$. Thus $r'_j = \sum_i k_{ji} r_i \geq 2(1 - \eta_n)r_j$. If $\eta_n \leq 1/4$, this implies $r'_j \geq 3r_j/2$, and hence $r'_j \leq 3|r'_j - r_j|$. The total μ_n -mass of all edges incident to such targets is consequently at most

$$\frac{3}{\widehat{d}_n} \sum_j s_j |r'_j - r_j| = o_{\mathcal{U}}(1).$$

If a source i is incident to two edges $(j_1, i), (j_2, i)$ of F_n , then similarly $s_{j_a} \geq s_i/(1 + \eta_n) \geq (1 - \eta_n)s_i$. Hence $s'_i = \sum_j k_{ji} s_j \geq 2(1 - \eta_n)s_i$ and $s'_i \leq 3|s'_i - s_i|$. The total mass of all edges incident to duplicated sources is at most

$$\frac{3}{\widehat{d}_n} \sum_i r_i |s'_i - s_i| = o_{\mathcal{U}}(1).$$

Remove from F_n all edges incident to a repeated source or target and call the remaining set E_n . It is a partial matching with $\mu_n(E_n) \rightarrow_{\mathcal{U}} 1$. Its vertices carry corrected source and target trace $1 - o_{\mathcal{U}}(1)$; total-variation convergence of the marginals gives the same conclusion for the original trace.

Let p_n be the sum of the occurrence spaces in E_n . Then $p_n \in \widehat{A}_n^+ \cap (\widehat{A}_n^-)'$ and $\widehat{\text{tr}}_n(p_n) \rightarrow_{\mathcal{U}} 1$. To verify the membership claim, fix a matched edge (j, i) . In target block j , its projection is the matrix-coordinate projection onto one $\mathbb{C}^{r_i}$ summand of $\mathbb{C}^{r'_j}$, so it belongs to $M_{r'_j} \otimes 1_{s_j}$. In source block i , it is the identity on $\mathbb{C}^{r_i}$ tensored with a multiplicity projection, so it commutes with $M_{r_i} \otimes 1_{s'_i}$.

Now let $a \in \widehat{A}_n^+$ be a contraction. On each matched edge (j, i) , compress the j -th target matrix of a to its selected $\mathbb{C}^{r_i}$ and use this as the i -th matrix block of an element $b \in \widehat{A}_n^-$. Put zero on unmatched source blocks. Since E_n is a matching, this defines b unambiguously, and $\|b\| \leq 1$. By construction, $p_n a p_n = p_n b p_n$. Since p_n commutes with b , we have

$$a - b = (1 - p_n)a + p_n a (1 - p_n) - (1 - p_n)b.$$

Each term on the right has normalized 2-norm at most $\widehat{\text{tr}}_n(1 - p_n)^{1/2}$. Hence

$$\|a - b\|_{2, \widehat{d}_n} \leq 3\widehat{\text{tr}}_n(1 - p_n)^{1/2} = o_{\mathcal{U}}(1).$$

We finally return to the original algebras. Let $a \in A_n^+$ be a contraction. Under the stable identifications in (3.4), choose a contraction $\widehat{a} \in \widehat{A}_n^+$ close to $W_n^+ a (W_n^+)^*$. The preceding paragraph gives a contraction $\widehat{b} \in \widehat{A}_n^-$ close to $\widehat{a}$, and (3.4) gives a contraction $b \in A_n^-$ such that $W_n^- b (W_n^-)^*$ is close to $\widehat{b}$. All these approximations are uniform over the respective unit balls. They take place on the retained corners; extending by zero over the discarded corners adds only $o_{\mathcal{U}}(1)$ because those corners have relative dimension tending to zero. On the stabilized space,

$$\begin{aligned} \|a - b\|_{2, \circ} &\leq 2\|W_n^+ - W_n^-\|_{2, \circ} + \|W_n^+ a (W_n^+)^* - \widehat{a}\|_{2, \circ} \\ &\quad + \|\widehat{a} - \widehat{b}\|_{2, \circ} + \|\widehat{b} - W_n^- b (W_n^-)^*\|_{2, \circ}. \end{aligned}$$

Every term tends to zero along $\mathcal{U}$; for the matching term we also use $\widehat{d}_n/d_n^{\circ} \rightarrow_{\mathcal{U}} 1$. Since $d_n^{\circ}/d_n \rightarrow_{\mathcal{U}} 1$, contractions in A_n^+ are uniformly at normalized 2-distance $o_{\mathcal{U}}(1)$ from contractions in A_n^- . This is the reverse near inclusion at finite level. Passing to the ultraproduct gives $A \subseteq u_{\ell}^* A u_{\ell}$. Together with the assumed inclusion this proves equality for every ℓ . $\square$

REMARK 4.2. The following standard example shows why a finite-level anchor is needed. Let

$$A_n = \bigoplus_{i=1}^n M_{2^i}(\mathbb{C}), \quad \tau_n(x_1, \dots, x_n) = \frac{1}{n} \sum_{i=1}^n \text{tr}_{2^i}(x_i),$$

so that the trace is distributed evenly among the n simple summands. This trace is the restriction of the normalized matrix trace under the realization

$$A_n = \bigoplus_{i=1}^n (M_{2^i}(\mathbb{C}) \otimes 1_{2^{n-i}}) \subseteq M_{n2^n}(\mathbb{C}).$$

Define

$$\theta_n(x_1, \dots, x_n) = (0, x_1 \otimes 1_2, x_2 \otimes 1_2, \dots, x_{n-1} \otimes 1_2).$$

Thus θ_n shifts every summand one place to the right, doubles its defining representation, and annihilates the final summand $M_{2^n}(\mathbb{C})$. Each θ_n is an exact nonunital, noninjective $*$ -homomorphism. For every contraction $x = (x_1, \dots, x_n)$,

$$\|1 - \theta_n(1)\|_{2, \tau_n}^2 = \frac{1}{n}, \quad |\tau_n(\theta_n(x)) - \tau_n(x)| \leq \frac{1}{n},$$

and

$$0 \leq \|x\|_{2, \tau_n}^2 - \|\theta_n(x)\|_{2, \tau_n}^2 = \frac{1}{n} \|x_n\|_{2, \text{tr}_{2^n}}^2 \leq \frac{1}{n}.$$

Consequently $(\theta_n)_n$ induces a unital trace-preserving injective $*$ -endomorphism θ of $\prod_{n \rightarrow \mathcal{U}} (A_n, \tau_n)$. This endomorphism is implemented by a unitary in the ambient tracial ultraproduct. Indeed, let $w_n \in U(n2^n)$ cyclically permute the n summand spaces of dimension 2^n , sending the i -th space to the $(i+1)$ -st and the last space to the first. Choose the identifications so that conjugation by w_n agrees with θ_n on the first $n-1$ input summands. For every contraction $x = (x_1, \dots, x_n) \in A_n$,

$$\|w_n x w_n^* - \theta_n(x)\|_2^2 = \frac{1}{n} \|x_n\|_{2, \text{tr}_{2^n}}^2 \leq \frac{1}{n}.$$

Hence $w = (w_n)_{n \rightarrow \mathcal{U}}$ implements θ in the ambient ultraproduct: $\theta(a) = waw^*$ for every $a \in \prod_{n \rightarrow \mathcal{U}} (A_n, \tau_n)$. It is not surjective. Indeed, set $v = \text{diag}(1, -1) \in M_2(\mathbb{C})$ and

$$y_n = (v, 1_2 \otimes v, 1_4 \otimes v, \dots, 1_{2^{n-1}} \otimes v) \in U(A_n).$$

In the first summand the range of θ_n is zero, while in the j -th summand, for $j \geq 2$, it is $M_{2^{j-1}}(\mathbb{C}) \otimes 1_2$. The trace-preserving conditional expectation of $1_{2^{j-1}} \otimes v$ onto this subalgebra is zero. Therefore $\text{dist}_{2, \tau_n}(y_n, \theta_n(A_n)) = 1$ for every n , and $(y_n)_{n \rightarrow \mathcal{U}}$ does not lie in the range of θ .

The lost input and output boundary summands each have trace $1/n$, so their defects disappear in the ultraproduct, while the proper matrix amplification persists in every summand. Thus asymptotic unitality and trace preservation alone do not prevent drift. It is likewise not enough merely to observe that each finite center has finitely many atoms. The internality of D , together with the identity $E_{D_n}(x_n) = \frac{1}{2}1$, provides the finite-level anchor that excludes this phenomenon and is essential to the argument. In the permutation proof of [17], the corresponding role is played by the ambient Kazhdan component decomposition and its weighted median.

REMARK 4.3. The distinction between genuine finite quotients and general sofic approximations is already visible in $BS(1, 2) = \langle a, t \mid tat^{-1} = a^2 \rangle$. In a height approximation, level i may be taken to consist of 2^{n-i} a -cycles of length 2^i , so every level has the same size. Away from the two end levels, t sends each cycle onto one of the two a^2 -orbits in a cycle at the next level. This is the permutation version of the shift in Remark 4.2. In contrast, in the finite quotients $C_{2^{n-1}} \rtimes C_n$, the generator of C_n acts by multiplication by 2, so $\langle a^2 \rangle = \langle a \rangle$ and there is no drift.

This behavior of finite quotients is not a consequence of the abelianity of $\langle a \rangle$. If $q: G \rightarrow Q$ is a homomorphism to a finite group and $tHt^{-1} \leq H$, then $q(t)q(H)q(t)^{-1} \leq q(H)$. The two groups have the same order, so they are equal. Thus a compressor always normalizes the finite image of H .

5. Proof of the main results

5.1. From the centralizer problem to normalization. We now assume a positive answer to the centralizer problem and prove Theorem 1.2.

PROPOSITION 5.1. *Let $H < G$ be Kazhdan groups and let $\pi: G \rightarrow U(\mathcal{M})$ be a homomorphism. Put $A = \pi(H)' \cap \mathcal{M}$ and $D = \pi(G)' \cap \mathcal{M}$. After stable negligible modifications, there are finite-dimensional algebras $D_n \subseteq A_n$ such that $A = \prod_{n \rightarrow \mathcal{U}} A_n$ and $D = \prod_{n \rightarrow \mathcal{U}} D_n$. If $t_1, \dots, t_m \in G$ and $u_\ell = \pi(t_\ell)$, representatives may be chosen so that $u_{\ell, n} \in D'_n$ for all n and ℓ .*

PROOF. Apply the centralizer problem to $\pi|_H$ and $\pi|_G$. Since both groups are Kazhdan, there are finite-dimensional algebras $D_n, A'_n \subseteq M_{d_n}(\mathbb{C})$ such that $A = \prod_{n \rightarrow \mathcal{U}} A'_n$ and $D = \prod_{n \rightarrow \mathcal{U}} D_n$.

Since $D \subseteq A$, Lemma 2.2 gives $D_n \subset_{2, \varepsilon_n} A'_n$ with $\varepsilon_n \rightarrow_{\mathcal{U}} 0$. Apply Proposition 3.2 to this near inclusion and use the common stable identification supplied by that proposition. Denote the corrected algebras by $D_n \subseteq A_n$; they still represent $D \subseteq A$.

Now choose unitary representatives $v_{\ell, n}$ of u_ℓ in the corrected ambient dimensions. Since $D \subseteq \{u_\ell\}'$, a diagonal argument gives $\sup_{d \in U(D_n)} \|[v_{\ell, n}, d]\|_2 \rightarrow_{\mathcal{U}} 0$.

Indeed, otherwise one could choose $d_n \in U(D_n)$ for which the commutators stay bounded away from zero; then $(d_n)_{n \rightarrow \mathcal{U}} \in D$ would not commute with u_ℓ . Haar averaging $v_{\ell,n}$ under conjugation by $U(D_n)$ gives an operator in D'_n at $o_2(1)$ -distance from $v_{\ell,n}$. Its polar part and support projections also lie in D'_n . Complete the polar part blockwise on its initial and final kernels to a unitary $u_{\ell,n} \in D'_n$. Since $v_{\ell,n}$ is unitary, the polar-decomposition estimate gives $\|u_{\ell,n} - v_{\ell,n}\|_2 \rightarrow_{\mathcal{U}} 0$. Thus the new sequence still represents u_ℓ . $\square$

PROOF OF THEOREM 1.2. The group G is finitely generated, and P_H generates G . Consequently there are $t_1, \dots, t_m \in P_H$ such that $G = \langle H, t_1, \dots, t_m \rangle$. Put $A = \pi(H)' \cap \mathcal{M}$ and $D = \pi(G)' \cap \mathcal{M}$. Apply Proposition 5.1 and choose representatives $u_{\ell,n} \in D'_n$ of $u_\ell = \pi(t_\ell)$. The choice of generators gives $D = A \cap \bigcap_{\ell=1}^m \{u_\ell\}'$. Thus Proposition 5.1 supplies all the hypotheses of Theorem 4.1 except the one-sided inclusions.

Let $x \in A$ and $h \in H$. Put $k = t_\ell h t_\ell^{-1} \in H$. Then $[u_\ell^* x u_\ell, \pi(h)] = u_\ell^* [x, \pi(k)] u_\ell = 0$. Thus $u_\ell^* A u_\ell \subseteq A$. By Theorem 4.1, equality holds for every ℓ . The algebra A is fixed pointwise by $\pi(H)$ and normalized by all u_ℓ , hence it is normalized by $\pi(G)$. $\square$

5.2. A conditional nonhyperlinear group. We finish the proof of Theorem 1.3.

PROOF OF THEOREM 1.3. Because H is infranormal but not normal, there is a strict compressor $t \in P_H$, that is, $tHt^{-1} < H$. Indeed, if every element of P_H normalized H , then the group generated by P_H would normalize H .

Write G_1 and G_2 for the two factors of P , and write g_i for the copy of $g \in G$ in G_i . Suppose that P is hyperlinear and choose an injective homomorphism $\pi: P \rightarrow U(\mathcal{M})$. Put $A = \pi(H)' \cap \mathcal{M}$. The element $c = t_1^{-1} t_2$ centralizes H : for $h \in H$, the element $t_2 h t_2^{-1}$ belongs to the common subgroup H and hence $t_1^{-1} (t_2 h t_2^{-1}) t_1 = h$. Thus $\pi(c) \in A$.

Apply Theorem 1.2 to the first copy G_1 . It follows that A is normalized by $\pi(G_1)$, and therefore $\pi(t_2 t_1^{-1}) = \pi(t_1 c t_1^{-1}) \in A$. Choose $h \in H \setminus tHt^{-1}$. The commutator $[h, t_2 t_1^{-1}]$ is nontrivial: after collecting letters from the same factor, its normal form is $(h t_2) (t_1^{-1} h^{-1} t_1) t_2^{-1}$. A strict compressor cannot belong to H , so the first and third letters lie outside H ; the middle letter lies outside H precisely because $h \notin tHt^{-1}$. The word is therefore reduced. On the other hand, $\pi(t_2 t_1^{-1}) \in A$ commutes with $\pi(h)$, so this nontrivial commutator lies in the kernel of π , a contradiction.

Kun and the author proved in [17, Theorem E] that, for every prime power q and $r, d \geq 3$, the group $H = \text{EL}_r(\mathbb{F}_q[x_1, \dots, x_d])$ and the group

$$G = \text{EL}_r(\mathbb{F}_q[x_1^{\pm 1}, \dots, x_d^{\pm 1}]) \rtimes \text{SL}_d(\mathbb{Z})$$

are residually finite Kazhdan groups and that H is infranormal but not normal in G . In fact, one may see a strict compressor explicitly. If $u_{12} = I + E_{12} \in \text{SL}_d(\mathbb{Z})$ and $t = (1, u_{12}) \in G$, then $tHt^{-1} = \text{EL}_r(\mathbb{F}_q[x_1, x_1 x_2, x_3, \dots, x_d]) < H$. For example, an elementary matrix with coefficient x_2 lies in H but not in tHt^{-1} . $\square$

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